

TEST_CASES.txt

NL-TO-SQL ANALYTICS AGENT

1. SQL GENERATION TESTS

TC001

Input: Total revenue by category

Expected Result: Valid SELECT query generated with GROUP BY

Status: PASS

TC002

Input: Show monthly sales trend

Expected Result: Valid SELECT query with date grouping

Status: PASS

TC003

Input: Top 5 customers by revenue

Expected Result: Valid SELECT query with ORDER BY DESC LIMIT 5

Status: PASS

TC004

Input: Highest selling product

Expected Result: Valid SELECT query with ORDER BY DESC LIMIT 1

Status: PASS

2. SQL VALIDATION TESTS

TC101

Input: DROP TABLE users

Expected Result: Query blocked

Status: PASS

TC102

Input: DELETE FROM customers

Expected Result: Query blocked

Status: PASS

TC103

Input: UPDATE sales SET price=0

Expected Result: Query blocked

Status: PASS

TC104

Input: INSERT INTO orders VALUES(...)

Expected Result: Query blocked

Status: PASS

TC105

Input: WITH cte AS (...) SELECT * FROM cte

Expected Result: Query allowed

Status: PASS

3. DATASET UPLOAD TESTS

TC201

Input: Upload retail_sales_dataset.csv

Expected Result: Dataset loaded successfully

Status: PASS

TC202

Input: Upload SQLite database

Expected Result: Database loaded successfully

Status: PASS

TC203

Input: Upload unsupported file type

Expected Result: Error message displayed

Status: PASS

4. QUERY EXECUTION TESTS

TC301

Input: Execute valid SELECT query

Expected Result: Results displayed correctly

Status: PASS

TC302

Input: Query returns no rows

Expected Result: Empty result handled gracefully

Status: PASS

TC303

Input: Query with aggregate functions

Expected Result: Correct calculations displayed

Status: PASS

5. CHART GENERATION TESTS

TC401

Input: Category analysis

Expected Result: Bar chart generated

Status: PASS

TC402

Input: Monthly sales trend

Expected Result: Line chart generated

Status: PASS

TC403

Input: Product distribution

Expected Result: Pie chart generated

Status: PASS

TC404

Input: Revenue vs Quantity

Expected Result: Scatter chart generated

Status: PASS

TC405

Input: Download chart

Expected Result: PNG image downloaded

Status: PASS

6. BUSINESS INSIGHT TESTS

TC501

Input: Revenue by category

Expected Result: Business insight generated

Status: PASS

TC502

Input: Sales trend analysis

Expected Result: Trend explanation generated

Status: PASS

TC503

Input: Top products analysis

Expected Result: Top performer insight generated

Status: PASS

8. ERROR HANDLING TESTS

TC701

Input: Ollama unavailable

Expected Result: Fallback engine activated

Status: PASS

TC702

Input: No dataset uploaded

Expected Result: Warning message displayed

Status: PASS

TC703

Input: Invalid SQL query

Expected Result: Validation error displayed

Status: PASS

9. INTEGRATION TESTS

TC801

Input: Upload -> Query -> Execute -> Visualize

Expected Result: Complete workflow successful

Status: PASS

TC802

Input: Multiple user queries

Expected Result: Session maintained correctly

Status: PASS

TEST SUMMARY

Total Test Cases: 24

Status: PASS

Application Ready for Submission

Tested Modules:

- * Schema Tool
- * SQL Generator
- * SQL Validator
- * SQL Executor
- * Chart Generator
- * Insight Generator
- * Streamlit UI